



Research paper

Spatial effect of digital economy on the coordinated development of Port cities

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ABSTRACT

Under the background of thriving digital economy and the digitization of port industry, using data derived from port cities in Guangdong, this study intends to investigate the spatial effect of digital economy on the port city coordination. This paper firstly measures the port city coupling coordination degrees and the digital economy comprehensive development levels, then estimates the spatial spill over effect of digital economy on the port city coordination by constructing the dynamic Spatial Durbin Model (SDM). To further clarify local patterns of the spatial effect, this paper subsequently analyses the spatial effects within and between three sub regions in Guangdong. The results show that first, the port city coordination has a habit persistence; second, the digital economy development may be mainly due to the local policy; third, the digital economy has a positive spatial spillover effect on the port city coordination, and the effect will increase over time; forth, the spatial effects in the Pearl River Delta and in the western and eastern areas present different patterns. The quantitative research enriches the port city theory under the new circumstances of digital economy, and brings new insights into regional integration and sustainable development.

1. Introduction

Integrating capital, technology, logistics and information, the port is the gateway and economic growth pole of the hinterland city and the regional economy (Dadashpoor & Arasteh, 2020). Policymakers not only focus on the economic effects of ports but also on their role of achieving balanced regional development, and upgrading industrial structure (WANG & DENG, 2020). Port city refers to a city of special type that has a port with the function of water and land transportation hub, and the port is the core advantage of city economic development (Deng et al., 2020). Port and city, located in a limited space, share resources and closely linked economically. Their coordinated development has been the foothold for both improving the comprehensive strength of port cities and realizing the regional sustainable development (Guo, Qin, Du, & Han, 2020).

With the deepening development of globalization, the coastal areas of developing countries are becoming carriers of global industrial transfer. Particularly Chinese ports are gradually integrated into the process, and the scale of ports are expanding. The China Port Yearbook reported that from 2001 to 2017, the cargo and container throughput of

Chinese ports have increased from 1.5 billion tons and 27 million TEU to 12.6 billion tons and 240 million TEU, with average annual growth rates of 14.2% and 14.6%, respectively (Deng et al., 2020; Ducruet, Cuyala, & El Hosni, 2018; Feng, Grifoll, & Zheng, 2019). On the other side of the rapid scale expanding, Chinese ports are suffering from a capacity surplus and serious homogenization, intensifying the vicious competition among ports, the result of which is the excessive waste of port resources (Chen & Song, 2019; Feng, Wang, Wen, Dong, & Chang, 2018; Hou, Long, Zhang, & Wu, 2020; Lam & Li, 2019). To overcome the vicious homogeneous competition and the subsequent uncoordinated development of port cities, it is very important to analyze the possible factors and their effects as part of the process of achieving sustainable regional economic development.

Meanwhile new content has been brought to the port city relationship context under the circumstances of digital economy. With the rapid development of information technologies, countries around the world are facing a profound information technology revolution (G, K, I, & G, 2021; Xu, Gao, & Huo, 2022). In this regard, the digital economy has become an important engine for global economic change and a driver for China's high-quality economic development. Recently, in 2023, the CPC

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Central Committee and the State Council issued the “Overall Layout Plan for Digital China Construction”, emphasizing the comprehensive improvement of the integrity, systematization and coordination of digital China construction, promoting the deep integration of the digital economy and the real economy, driving the reform of production, life and governance with digitization. At present, featured by the integration of port business and technology which is represented by the Internet of Things (IOT), big data, cloud computing, and artificial intelligence (AI), smart ports have become the focus of a new round of global competition (Li et al., 2023). China has launched the construction of smart port demonstration projects in 2017, and the integration of digital economy and port business as well as the integration of digital economy and urban industry is bound to influence the port city coordination.

In particular, Guangdong is a typical place for research which relates port city coordination and digital economy simultaneously. For Guangdong province ranks top in China in terms of digital economy scale, with the digital economy accounting for more than 50% of the Gross Domestic Product (GDP) in 2021. Also as a strong province in multimodal transportation, Guangdong attaches more importance to the port city coordinated development (Tang & Lin, 2023). The port layout plan of Guangdong Province (2021–2035) puts forward the guidance of the construction of a strong communication province, taking the lead in building a world-class port in China, and building a “one core and two poles” development pattern with the Pearl River Delta as the core, the eastern and western areas as the development poles. To achieve the goal of building a world-class port cluster, it is necessary to adhere to the digital economy as a drive to port cities.

Among the possible effects which digital economy acts on the port city coordination, spatial effect is from a unique perspective. As regional integration progresses, growth in one region's digital economy and port industry may have a spillover effect on the development of other regions. In terms of resource elements, the low diffusion cost and high diffusion speed of data cause them to flow naturally. This type of mobility is less limited by time and space, leading to a significant geographical spillover effect (He, Sun, Gao, & Lau, 2019; Li, Shi, & Tian, 2018; Zhao, Zhang, & Liang, 2020). In addition, China has encouraged the development of regional integration, strengthening the economic linkages between regions. Besides, considering the “Metcalf rule” of the Internet, the network spillover shows the characteristics of increasing marginal effect. Therefore, the possible impact of digital economy on port city coordinated development is likely to have non-linear characteristics and spatial spillover effects on neighboring regions (Zhao et al., 2020).

Hence this study intends to find empirical evidence about the spatial effects of digital economy on the port city coordination to enrich the port city theory under the new era of modernization. The quantitative data is derived from port cities in Guangdong from 2011 to 2019, a typical area where ports are densely allocated and the digital economy is vigorously developing.

The structure of this paper is presented as follows. The first part is an introduction, which briefly introduces the reasons for choosing the topic and the significance of the research. The second part is a literature review, which provides a comprehensive overview of scholars' research results on the topic, and presents the innovative points of this paper. The third part is the research design and data selection, showing the measurement indicators, research methodology, data sources and model construction. The fourth part reports the empirical results, analyzing the time-series changes, regional differences, spatial effects and other driving factors of the digital economy on the port city coordination. The fifth part is the discussion of empirical investigation, results and policy recommendations. The sixth part is the conclusions and reflections on the limitations for improvement of the paper.

2. Literature review

Scholars have made unremitting research and exploration on the

interaction mechanism and key factors of port and its hinterland city. Most of the studies focus on the interaction of port indicators and hinterland economic development, such as port scale (Guo et al., 2020; A, S, S, & R, 2021), port freight infrastructure (Liu, Schindler, & Liu, 2020), port operation (Han, Wang, & Li, 2019; Lim, Pettit, Abouarghoub, & Beresford, 2019; Mudronja, Jugovi, & Skalamera-Alilovi, 2020); some of the studies utilize econometrics to explore the interaction mechanism of port city economy (Cong, Zhang, Wang, Xu, & Li, 2020; Lu, Xing, Wang, & Wang, 2019), and others calculate and analyze the coupling coordination relationship of port and city (WANG & DENG, 2020). At the same time, scholars also found that when the port city coordination is insufficient, the development and space expansion of both port and city will bring mutual interference and restrictions (Guo et al., 2022, Sdoukopoulos & Boile, 2020).

To quantitatively measure the coordinated development of the port and city economy, scholars usually use statistical or System Dynamics methods. The statistical methods mainly adopt the statistics indicators of the relative development degree of port and city, such as gray correlation degree, relative concentration index (RCI), Gini coefficient, etc (Kong et al., 2021; Guo et al., 2020; Lin & Tang, 2023). And the System Dynamics methods recently apply the coupling coordination degree model based on self-organizing theory (Lu, Qiu, Xing, & Wen, 2020, WANG & DENG, 2020). RCI, Gini coefficient and other statistics indicators are simple and direct for the determination of the coordinated development of port and city, but the grade division is highly subjective. Comparatively the coupling coordination model based on the complex index system can reveal the internal discipline of the interactive development of port and city in depth. Besides, representative researches also reveal the factors which affect the coordinated development of port city, those factors mainly including port infrastructure, port operation level, port service function, foreign trade scale, urban economic scale, industrial structure, distribution conditions, government attention and policy support, etc (Lu et al., 2019; Cong et al., 2020, Wang & Deng, 2020; Guo & Qin, 2022).

In addition to the traditional factors, given the explosive growth of the digital economy, the discussion of the impact of the digital economy on the smart ports has begun to emerge (Lee, Lam, Lin, Hu, & Cheong, 2018). Just as smart city is a typical model of urban development in the digital age, smart port is also a new concept put forward by the integration of digital economy and port industry, and can promote the implementation of port function, technology and service innovation on the basis of information processing and sharing (Li et al., 2023). As president Xi Jinping has pointed out that new drivers should be fostered to promote new economy, the digital economy will also release new momentum to the port operation and the port city coordination development as well. The digital economy may affect the port city coordination in several ways. Firstly, the digital economy prompts industrial digitization and facilitate industrial integration through digital elements, which help to bring about new business formats or services, such as the cross-border electricity, digital trade etc. Hence it greatly improves the comprehensive service level of the port, integrate resources, transforms the port function, and extends the port logistics services, thereby strengthen the linkage of the port city (Shi D., 2022). Secondly, the digital economy reduces transaction costs, improves market potential, and realizes the scale of economy, generating capital and human resources agglomeration. Capital agglomeration can expand the multiplier effect of ports on urban economic growth, while human agglomeration can directly promote innovation in the port and the urban area thus bring about rapid growth (Xu et al., 2022). Moreover, the Internet technology supporting the digital economy has broken the spatial distance between the port and the city, and is reshaping the economic and geographical location potential, and is becoming an important strategic technology for the regional coordinated development (G D., 2021). Finally, the new digital infrastructure promotes the upgrading of ports and urban enterprises, and improves the production efficiency of urban enterprises and port operational efficiency, so as to promote the

high-quality coordinated development of port cities (A G., 2021).

Although the integration of digital economy and port industry has received huge policy emphasis, existing literature simply focus on the role of digital economy on port or on city separately, and little attention has been devoted to the impact of digital economy directly on the coordinated development of port-city complex system. In addition, even fewer studies have explored the possible spatial spillover effects of digital economy on the port city coordination. However, studies have found that with the improvement of port capacity and the expansion of radiation range, ports will exert spillover effects on hinterland cities (Deng et al., 2020). Besides, spatial effects have been shown to be an important cause of sequence cross-sectional correlation in the application of econometric methods, and ignoring spatial spillover effects can bias the quantitative estimation of the port city interaction (Deng et al., 2020; Deng, Bai, Shen, & Xia, 2022). Furthermore, since the spatial effect of digital economy on regional economy is also significant (Pan, Xie, W, & Ma, 2022; Cheng, Zhang, Wang, & Jiang, 2023), the impact of digital economy on the port city coordination should also have a spatial effect theoretically.

In view of this, the possible marginal contribution of this paper are as follows. First, this study enriches the port city relationship theory by providing empirical research of the influence of the digital economy, a new driving factor, on the port city coordination. Second, this study depicts the spatial characteristics of the port city coupling coordination degree and the digital economy development level in Guangdong using spatial statistics. Last, this study establishes a connection of digital economy and port city coordination by providing the tests of dynamic spatial Durbin econometric model, from the perspective of spatial distance effects.

3. Research design and data collection

In order to explore the influence of digital economy on the coordinated development of port cities from the perspective of spatial effects, by collecting panel data from Guangdong port cities, this paper first constructs the coupling coordination degree model to measure the port city coordinated development, then uses the research idea of Zhao et al. (2020) and takes a TOPSIS entropy method to measure the comprehensive digital economy development of port cities, finally on the basis of the spatial correlation analysis of both coupling coordination degrees and digital economy development, a spatial econometric model is built to explore the spatial effects of digital economy development on the coupling coordination degrees of port cities.

3.1. Port city coupling coordination degree measurement

Based on the order parameter and slaving principles of Synergetics, the coupling coordination degree model has been widely used in physics and economics to study multi-system interactions. Since the port and city are generally treated as a nested system which is composed of port subsystem and city subsystem (Kong et al., 2021), this paper constructs a coupling coordination degree model to measure the coordinated development degrees of port cities. Of the port-city complex system, suppose the port subsystem is S_1 , and the hinterland economic subsystem is S_2 . With subsystem S_i $i = [1, 2]$, the order parameters in the coordination evolution process are $e_i = (e_{i1}, e_{i2}, \dots, e_{in})$, $n \geq 1$. Each certain sequence parameter e_{ij} has m samples, and satisfies $U_{ij} \leq e_{ij} \leq T_{ij}$, where T_{ij} is the upper and U_{ij} is the lower limit of e_{ij} . The order degree of a subsystem is calculated by formula (1):

$$\mu(e_{ij}) = \begin{cases} \frac{e_{ij} - U_{ij}}{T_{ij} - U_{ij}}, j \in (1, k_1) \\ \frac{T_{ij} - e_{ij}}{T_{ij} - U_{ij}}, j \in (k_1 + 1, n) \end{cases} \quad (1)$$

Obviously, $\mu(e_{ij}) \in [0, 1]$, and the greater its value, the higher the

contribution of e_{ij} is to the order of the subsystem.

$$\mu_i = \sum_{j=1}^n \omega_j \mu(e_{ij}), \omega_j \geq 0 \cap \sum_{j=1}^n \omega_j = 1 \quad (2)$$

The total contribution of all parameters e_{ij} to subsystem S_i is calculated by formula (2). The weight coefficient ω_{ij} is for the sequence parameter e_{ij} , which can be determined by the entropy weighting method. Furthermore, Formula (3) introduces the coupling degree function in physics to calculate the coupling degree of the subsystems.

$$C = \left[\frac{\prod_{i=1}^2 \mu_i(e_i)}{\left(\frac{\sum_{i=1}^2 \mu_i(e_i)}{2} \right)^2} \right]^{1/2} \quad (3)$$

The coupling degree C falls into the range of $[0, 1]$. The larger the value, the higher the coupling degree between the subsystems. Based on the coupling degree C of subsystems, in order to reflect the coordinated development of the subsystems, the coupling coordination degree CCD is introduced.

$$CCD = \sqrt{C \times D} \quad (4)$$

$$D = \left(\frac{\sum_{i=1}^2 \mu(e_i)}{2} \right) / 2 \quad (5)$$

Similar to the coupling degree C , the coupling coordination degree CCD is also between 0 and 1, and can be divided into four levels by using the mid-segment method (Liu et al., 2018, Kong & Liu, 2021), such as in Table 1.

3.2. Digital economy development measurement

At present, there are few studies involving specific measurement of digital economy. Drawing on Zhao et al. (2020), taking Internet development as the core of measurement and adding the construction of digital transaction indicator system, this paper, in combination with the availability of relevant data at the city level, measures the comprehensive development level of digital economy from two aspects: internet development and digital inclusive finance. The internet development is characterized by four indicators: Internet penetration rate, relevant

Table 1
Classification of coupling coordination degree.

CCD Value	Coordination level	Characteristics
(0–0.3]	Low coupling stage	The economic development of the port city is low and backward, and the interaction between port and city is weak.
(0.3–0.5]	Antagonism stage	The economic development of the port city is gradually improving, along with the upgrading of ports. Thus the mutual promotion between the port and city has been strengthened.
(0.5–0.8]	Running-in stage	The two subsystems have reached the middle and high levels, developed steadily through interaction, constantly running in and realized orderly development.
(0.8–1]	High coupling stage	The degree of coupling coordination is high, the city economy is conducive to promoting the development of ports, while ports can well meet the development of cities in logistics, foreign trade, and other aspects, both subsystems are at a stable and high development level.

employees, relevant output and mobile phone penetration rate. The digital inclusive finance development is characterized by the China Digital Inclusive Finance Index.¹ To effectively use all these indicators, TOPSIS entropy weight method is applied to reduce the dimensionality.

When determining weight, The TOPSIS model is quite subjective, whereas the entropy method is not. The specific steps of the entropy TOPSIS algorithm are as follows (Deng et al., 2020; Li et al., 2018).

1. Use the range method to standardize the data to abandon the magnitude and dimensional differences in different indicators;
2. Calculate the information entropy H_j of evaluation index b_{ij} in the index set of digital economy.
3. Calculate the weight W_j of evaluation index b_{ij} in the index set of digital economy.
4. Construct the weight matrix D of each evaluation index.
5. Determine the positive ideal solutions T^+ and negative ideal solutions T^- of each index according to the weight matrix D gained by the entropy weight.
6. Calculate the distance between positive ideal solutions D^+ and the evaluation object and the distance between negative ideal solutions D^- and the evaluation object.
7. Calculate the relative adjacent degree C between the ideal solutions and the evaluation object in the digital economy index set.

$$C_i = \frac{D_i^+}{D_i^+ + D_i^-} \quad (i = 1, 2, \dots, m) \quad (6)$$

Where $C_j \in [0, 1]$, and the evaluation objects are sorted on the basis of their relative adjacent degree; thus, the larger C_j is, the closer the evaluation objects are to the ideal solutions.

3.3. Indicators and data collection

In order to estimate the plausible spatial effect, this paper constructs a spatial econometric model with port city coupling coordination degree as the dependent variable and the digital economy development level as the main independent variable.

1. Port city coupling coordination degree ($\text{port_city_coord}_{it}$), calculated by the coupling coordination degree model, representing the level of coordinated development of port cities. The calculation index set includes:
 - the cargo throughput, container throughput describing the port operation.
 - port berth length, the number of berths, and the number of berths over 10,000 tons reflecting the level of port infrastructure construction.
 - the GDP, Import & Export representing the economic and trade condition in the hinterland city.
2. Digital economy development level (deconomy_{it}), the main independent variable, which describes the comprehensive development level of the digital economy in a place. The calculation index set includes:
 - Internet penetration rate, the number of Internet broadband access users in 100 people.
 - Relevant employees, the proportion of employees in computer service and software industry in urban units.
 - Telecom business per capita.
 - Postal business per capita
 - Mobile phone penetration rate, number of mobile phone users in 100 people

The above five indicators together interpret internet development, and the below indicator illustrates the digital inclusive finance development.

- the China Digital Inclusive Finance Index.

3. Control variables, composed of other factors that may affect the port city coordinated development:
 - Urban economic growth level (lnpgdp_{it}), which takes the average GDP per capita to represent the economic growth level of a hinterland city;
 - Port container throughput (Incontainer_{it}), which takes the container throughput to represent the port operation and logistics scale;
 - Economic and industrial structure ($\text{Industrystruc}_{it}$), calculated as the ratio of the added value of the tertiary industry to the added value of the secondary industry (Zhao et al., 2020), reflecting the upgrading of the industrial structure;
 - Infrastructure investment (fassetp_{it}), which is the ratio of fixed asset investment to GDP of a port city.

For data collection, this paper uses the annual panel data of representative port cities in Guangdong from 2011 to 2019. The data comes from the China Port Yearbook over the years, and the statistical yearbooks of Guangdong cities. The interpolation method is used to fill some individual missing data. In order to reduce the heteroscedasticity and unsteadiness, the original data are standardized and some of the indicators logarithmically processed. All price related data in the variables are deflated based on the CPI in 2005. The descriptive statistics of the selected data are reported in Table 2.

3.4. Spatial model construction

Before construct the spatial econometrics model, this paper checks the spatial autocorrelation effects of both the digital economy development levels and the port city coupling coordination degrees.

3.4.1. Spatial weight matrix and spatial correlation test

In order to improve the robustness of the spatial statistical results, this paper constructs two spatial weight matrices in calculation: reciprocal distance weight matrix and economic weight matrix.

The reciprocal distance weight matrix is composed as the following:

$$W_d = \begin{cases} 0, & i = j \\ \frac{1}{d_{ij}}, & i \neq j \end{cases} \quad (7)$$

The spatial weight matrix is $W = (w_{ij})_{n \times n}$. n is the number of port cities involved, and w_{ij} represents the spatial association between port i and port j . d_{ij} indicates the actual geographical distance between port i and port j . The economic weight matrix is constructed as follows:

$$W_e = \begin{cases} 0, & i = j \\ \frac{1}{|m_i - m_j|}, & i \neq j \end{cases} \quad (8)$$

m_i refers to the economic strength of port i , measured by the average per capital real GDP during the sample period.

To check and measure the spatial autocorrelation, Moran's I index is generally used as a representative statistic. Particularly, global Moran's I is able to reflect the overall level of spatial autocorrelation of observed objects. Moran's I index is defined as in formula (9):

¹ The China Digital Inclusive Finance Index is jointly prepared by the digital finance research center of Peking University and Ant financial service group.

Table 2
Descriptive statistics.

	Variables	Mean	Standard Deviation	Min	Max
Port city coupling coordination calculation	cargo_throughput	10256.04	12878.95	489	62687
	container_throughput	339.46	683.21	0	2576
	berth_lenth	15067.77	16178.88	1286	70999
	the number of berths	162.88	196.13	7	870
	the number of berths over 10,000 tons	17.1	23.26	0	78
	GDP	4044.48	5598.07	404.29	26992.33
Comprehensive digital economy development level calculation	Import & Export	3645.00	6991.42	49.40	33286.84
	Internet penetration rate	29.35	21.97	3.70	139.68
	Relevant employees	1.49	1.00	0.33	6.26
	Telecom business per capita	1744.96	1952.95	63.02	14777.79
	Postal business per capital	607.61	1022.18	20.33	7108.38
	Mobile phone penetration rate	1.33	1.05	0.42	9.79
Control variables	China Digital Inclusive Finance Index	175.86	65.94	41.63	306.67
	lnpgdp	10.71	0.61	9.53	11.98
	industrystruc	1.00	0.38	0.56	2.63
	fassetp	0.4753	0.3163	3.5241	0.1618
	lncontainer	3.79	2.77	-2.30	7.85

$$I = \frac{\sum_{i=1}^n \sum_{j=1}^n w_{ij} (x_i - \bar{x})(x_j - \bar{x})}{S^2 \sum_{i=1}^n \sum_{j=1}^n w_{ij}} \quad (9)$$

x_i refers to the indicator upon which the spatial autocorrelation effect is tested. $\bar{x} = \frac{\sum_{i=1}^n x_i}{n}$, $S^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n}$. Moran's I index is between -1 and 1. This paper uses two kinds of spatial matrices to check the spatial autocorrelation of both digital economy levels and port city coupling coordination degrees. The spatial correlation results are reported in Table 7.

3.4.2. Series stationary test

In order to avoid spurious regression, unit root tests are conducted to examine whether the variable series are stationary. Panel unit root tests generally include the first generation unit root tests (LLC test, LPS tests) and the second generation unit root tests (Fisher-ADF and Fisher-PP). In this paper, the commonly used Fisher-ADF test is used and the test results are shown in Table 3.

Table 3 shows that, given 5% significance level, the port city coupling coordination degree ($port_city_coord_{it}$), the comprehensive development level of digital economy ($deconomy_{it}$), the economic and industrial structure ($industrystruc_{it}$), the fixed asset investment proportion ($fassetp_{it}$), the container throughput ($lncontainer_{it}$), and the logarithmic per capital GDP ($lnpgdp_{it}$) are stationary.

Table 3
Unit root test result.

Variables	Indication	Fisher-ADF With time trend	Fisher-ADF Without time trend
$port_city_coord_{it}$	Port city coupling coordination degree	10.884*** (0.0000)	3.6407*** (0.0001)
$deconomy_{it}$	Comprehensive development level of digital economy	19.1541*** (0.0000)	2.1949** (0.0141)
$lnpgdp_{it}$	Per capital Gross Domestic Product (GDP)	14.0690*** (0.0000)	13.9480*** (0.0000)
$industrystruc_{it}$	Economic and industrial structure	5.1863*** (0.0000)	5.5903*** (0.0000)
$fassetp_{it}$	Fixed asset investment proportion	0.6017 (0.2737)	1.7556** (0.0396)
$lncontainer_{it}$	Container throughput	6.2891*** (0.0000)	17.9848*** (0.0000)

Note: The numbers in brackets are P values; ***, **, * represents the rejection of the null hypothesis of the existence of unit roots at the significant level of 10%, 5%, and 1%, respectively.

3.4.3. Spatial model setting inspection

The spatial model setting inspection is to be conducted before a proper spatial econometric model is build. According to ELHORST (2014), the general spatial model test judgment framework consists of Lagrange Multiplier (LM) test of Spatial Autoregression Model (SAR), Spatial Errors Model (SEM), Spatial Autocorrelation Model (SAC), Hausman test of fixed effects and random effects of SDM model, and Wald test of SDM model simplified into SAC, SAR and SEM, which reflects the test idea of combining "specific to general" and "general to specific".

Therefore the LM tests, Hausman tests and Wald tests are carried out in turn. The model setting inspection results are reported in Table 4.

Both the LM and the Wald test results show that SDM is better than SAR, SEM and SAC models under both the reciprocal distance and the economic weight matrices. In addition, for the two spatial weight matrices, the fixed effects, including time and individual, are superior to the random effects model in the Hausman tests.

3.4.4. Spatial econometric model setting

According to the model setting tests results in Tables 4 and in order to control the impact of the temporal lag term of the port city coupling coordination degree at the same time, the dynamic SDM model with dual fixed effects both in time and space is determined as the best choice. Referring to the viewpoint of port economy and the general anyport-type model, apart from the spatialtemporal lag terms of digital economy and the port city coupling coordination degree which are necessary to form a dynamic SDM model, some other control variables are also included. The dynamic SDM model are given in formula (10).

$$\begin{aligned}
 port_city_coord_{it} = & \tau port_city_coord_{it-1} + \rho \sum_{j=1}^n w_{ij} port_city_coord_{jt} \\
 & + \eta \sum_{j=1}^n w_{ij} port_city_coord_{jt-1} + \beta_1 deconomy_{it} + \theta_1 \sum_{j=1}^n w_{ij} deconomy_{jt} \\
 & + \beta_2 Z_{it} + \theta_2 \sum_{j=1}^n w_{ij} Z_{jt} + v_i + v_t + \varepsilon_{it}
 \end{aligned} \quad (10)$$

where i and t represent the port and year respectively; $port_city_coord_{it}$ represents the port city coupling coordination degree of the i th port in year t , which is the explained variable; $port_city_coord_{it-1}$ is the temporal lag term of the port city coupling coordination degree, τ is the coefficient of the temporal lag term, measuring the impact of the past port city coordination. w_{ij} is the element of spatial weight matrix W . $\sum_{j=1}^n w_{ij} port_city_coord_{jt}$ is the spatial lag term of the port city coupling coordination degree, and ρ is the spatial autoregression coefficient, which measures the impact of the port city coordination in the

Table 4

Test results of model selection.

Test method	Reciprocal distance weight matrix (Wd)	Economic weight matrix (We)	Test method	Reciprocal distance weight matrix (Wd)	Economic weight matrix (We)
LM Error (Burrige)	52.8542*** (0.0000)	11.8265*** (0.0006)	Wald test of SAR	27.96*** (0.0000)	51.37*** (0.0000)
LM Error (Robust)	0.0464 (0.8294)	8.5436*** (0.0035)	Wald test of SEM	19.04*** (0.0019)	52.37*** (0.0000)
LM Lag (Anselin)	61.3219*** (0.0000)	20.2403*** (0.0000)	Hausman test (time fixed VS random)	749.17*** (0.0000)	451.11*** (0.0000)
LM Lag (Robust)	8.5141*** (0.0035)	16.9214*** (0.0000)	Hausman test (individual fixed VS random)	136.66*** (0.0000)	102.79*** (0.0000)

Note: P value in brackets, * means significant at 10% level, ** means significant at 5% level, *** means significant at 1% level.

neighborhood. $\sum_{j=1}^n w_{ij}port_city_coord_{jt-1}$ is the spatiotemporal lag term of the port city coupling coordination degree, with η as the coefficient. $deconomy_{it}$ represents the digital economy development level of the i th port in year t , with β_1 as the coefficient. $\sum_{j=1}^n w_{ij}deconomy_{jt}$ represents the spatial lag term of the development level of the digital economy with θ_1 as the coefficient. Z_{it} is the control variable set, representing other factors that affect the port city coupling coordination degree with β as the coefficient set. $\sum_{j=1}^n w_{ij}Z_{jt}$ represents the spatial lag terms of the control variables, with θ standing for the coefficient set. v_i refers to the individual fixed effects of each port that does not change with time. v_t controls the fixed time effects. ε_{it} is a random error term.

4. Empirical results analysis

4.1. Port city coupling coordination degrees analysis

The calculated port city coupling coordination degrees of 16 port cities in Guangdong are reported in [Tables 5-1](#) and [Table 5-2](#). Specifically, [Tables 5-1](#) reports the ports in the Pearl River Delta, and [Tables 5-2](#) reports the ports in eastern and western Guangdong areas.

[Tables 5-1](#) and [Table 5-2](#) show the development of coupling coordination degrees of Guangdong port cities from 2011 to 2019. It can be seen that the coupling coordination degrees of different port cities in Guangdong are with distinct characteristics, and the heterogeneity can be observed more clearly in [Fig. 1](#) below.

[Fig. 1](#) shows the tendency of the coupling coordination degrees in Guangdong based on the calculation results in [Tables 5-1](#) and [Tables 5-2](#). The left chart of [Fig. 1](#) shows the trend of coordinated development of port cities in the Pearl River Delta, while the chart on the right shows those in eastern and western Guangdong areas. In general, the coordinated development of port cities in the Pearl River Delta is better than the port city coordination in eastern and western areas.

For the Pearl River Delta, the coastal port city coordination is better than the coordinated development of inland river ports. Among the coastal port cities in Pearl River Delta, Dongguan developed the most rapidly from 2011 to 2019, and became the one with the highest level of coordination in 2019, although it started low in 2011. Among the inland river ports in the Pearl River Delta, Foshan has the most stable port city coordination development.

As to the port clusters in eastern and western Guangdong, Zhanjiang has developed the best port city coordination in the western area, and Shantou has developed the best in the eastern area. These two port cities have been far ahead in their respective regions, indicating core port position. In the eastern area, Shanwei also has shown good performance, whereas other port cities have been mediocre, especially Chaozhou and Jieyang, which are costal ports, but with port city coupling coordination degrees similar to the inland river ports.

In general, the port city coupling coordination degrees in Guangdong have shown an upward trend, but there are also exceptions, such as Zhongshan and Zhaoqing in the Pearl River Delta region, as well as Chaozhou and Jieyang in eastern area. Zhongshan and Zhaoqing are inland river ports. For Chaozhou, a coastal port, the persisting declining port city coupling coordination degree may due to the continuously shrinking in imports and exports.

In more detail, the coupling coordination degrees of Guangdong port cities by 2019 can be classified into four categories according to [Table 1](#).

1. High coordination stage. Dongguan, Shenzhen, Jiangmen, Guangzhou, Zhuhai and Huizhou are in this stage. With superior geographical positions and abundant port resources. Guangzhou and Shenzhen have been among the world's top 10 container ports. In Dongguan, the annual container throughput was 50 million TEU in 2010 and 405 million TEU in 2019, showing an average annually increase of over 24%. In Zhuhai, the annual container throughput was 70 million TEU in 2010 and 256 million TEU in 2019, showing an average annual increase of over 11%. In Huizhou, the annual container throughput was 27 million TEU in 2010 and 41 million TEU in 2019. In terms of GDP, Shenzhen, Guangzhou, Dongguan and Huizhou are among the top five cities in Guangdong. Shenzhen's GDP in 2019 reached 2692.71 billion yuan, reaching a growth rate of 6.7%. In 2019, the GDP of Guangzhou was 2362.86 billion yuan, an increase of 6.8% over the previous year. Zhuhai and Jiangmen have not achieved so rapid economic growth, but their development between port and city is synchronized.
2. Running-in stage. This stage is a medium coordination level. The cities at this stage include Foshan, Shanwei and Maoming. Foshan is an inland port. Although Foshan is among the top 10 inland ports in China, as an inland port, it is inferior to seaports in the port city

Table 5-1

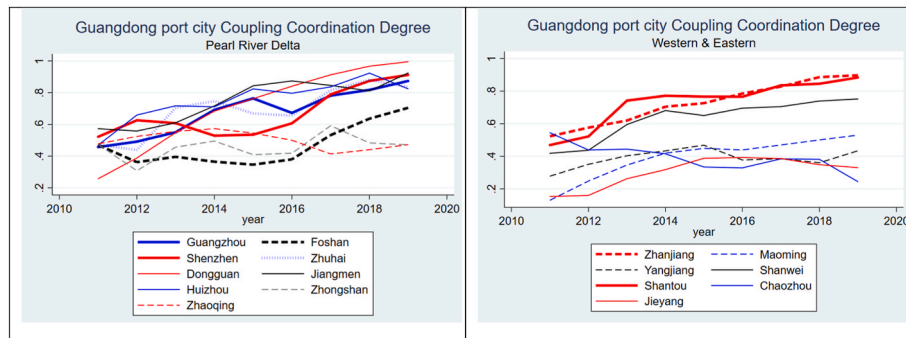
Port city coupling coordination degrees in Guangdong (Pear River Delta).

Year	Guangzhou	Foshan	Shenzhen	Zhuhai	Dongguan	Jiangmen	Huizhou	Zhongshan	Zhaoqing
2011	0.4572	0.4701	0.5223	0.4715	0.2577	0.5732	0.4689	0.4748	0.4773
2012	0.4913	0.3621	0.6263	0.4383	0.3881	0.5585	0.6593	0.3081	0.5244
2013	0.5507	0.3956	0.6080	0.7051	0.5481	0.6104	0.7178	0.4563	0.5564
2014	0.6921	0.3650	0.5295	0.7473	0.6840	0.7166	0.7113	0.4951	0.5734
2015	0.7642	0.3464	0.5357	0.6694	0.7624	0.8432	0.8237	0.4093	0.5459
2016	0.6729	0.3798	0.6070	0.6558	0.8409	0.8741	0.7962	0.4183	0.4999
2017	0.7816	0.5331	0.7893	0.8161	0.9129	0.8461	0.8367	0.5923	0.4137
2018	0.8177	0.6364	0.8744	0.8813	0.9670	0.8110	0.9235	0.4832	0.4407
2019	0.8741	0.7050	0.9122	0.8354	0.9951	0.9244	0.8242	0.4710	0.4724

Table 5-2

Port city coupling coordination degree in Guangdong (western and eastern areas).

Year	Zhanjiang	Maoming	Yangjiang	Shanwei	Shantou	Chaozhou	Jieyang
2011	0.5232	0.1297	0.2778	0.4181	0.4685	0.5448	0.1529
2012	0.5759	0.2478	0.3505	0.4366	0.5218	0.4382	0.1595
2013	0.6196	0.3459	0.4029	0.5951	0.7418	0.4437	0.2623
2014	0.7043	0.4190	0.4328	0.6807	0.7717	0.4152	0.3179
2015	0.7261	0.4482	0.4679	0.6505	0.7662	0.3340	0.3876
2016	0.7854	0.4389	0.3771	0.6956	0.7655	0.3287	0.3921
2017	0.8278	0.4691	0.3850	0.7050	0.8351	0.3840	0.3859
2018	0.8859	0.5000	0.3598	0.7392	0.8453	0.3809	0.3481
2019	0.8976	0.5300	0.4336	0.7519	0.8837	0.2441	0.3302

**Fig. 1.** The coupling coordination degrees in Guangdong.

coordinated development. Limited to water depth conditions, Foshan does not have berths above 10,000 tons. Thus compared to its city economy development level, its port production is rather lagged behind. Located in eastern Guangdong, Shanwei has abundant costal resources, but it's development is in progress, especially in its container throughput. Shanwei has not attracted much container transport business yet, thus cannot compete its counterpart in the Pearl River Delta. Maoming is an inland port in the western Guangdong area where is comparatively less developed. Maoming is not only limited to inferior water depth sources, but also enslaved to less developed hinterland economy.

3. Antagonism stage. Zhongshan, Yangjiang, Jieyang, Zhaoqing are in this stage. Among them, Zhongshan and Zhaoqing are inland ports in the Pearl River Delta. In Zhongshan, the import export trade has a declining tendency during the 9 years. Yangjiang is a coastal port in the western Guangdong area, and Jieyang is a coastal port in the eastern Guangdong area.
4. Low coordination stage. Only Chaozhou is in this stage. Since 2017, due to the influence of domestic and foreign economic situation, the cargo throughput in Chaozhou has begun to decline. In 2017, the cargo throughput of Chaozhou Port was 38.292 million tons, down by 3.9% from 2016. In 2019, the cargo throughput of Chaozhou further decreased by about 16.2%. In line with the decline in cargo throughput, so did the import and export trade. Nevertheless, the urban development of Chaozhou is relatively less affected.

Therefore, the development of the port and city is not synchronized, and the coordination degree of the port city was decreasing year by year.

4.2. Comprehensive digital economy development level analysis

The calculated comprehensive digital economy development levels of 16 port cities in Guangdong are reported in [Tables 6-1](#) and [Table 6-2](#). Specifically, [Tables 6-1](#) reports the port cities in the Pearl River Delta, and [Tables 6-2](#) reports the port cities in eastern and western Guangdong areas.

As shown in [Tables 6-1](#) and [Table 6-2](#), the comprehensive development level of digital economy of all port cities in Guangdong have increased from 2011 to 2019. The general tendency can be observed more clearly in [Fig. 2](#) below.

[Fig. 2](#) shows the comprehensive digital economy development in Guangdong based on the calculation results in [Tables 6-1](#) and [Tables 6-2](#). The left chart of [Fig. 2](#) shows the trend of the comprehensive digital economy development in the Pearl River Delta, while the chart on the right shows the trend in eastern and western Guangdong areas. In general, the comprehensive digital economy development of port cities in the Pearl River Delta is better than that in eastern and western areas. In western Guangdong, the digital development levels keep almost the same pace. Whereas Jieyang increases rapidly and the other port cities remained stagnant in eastern Guangdong. Among these cities, Shenzhen,

Table 6-1

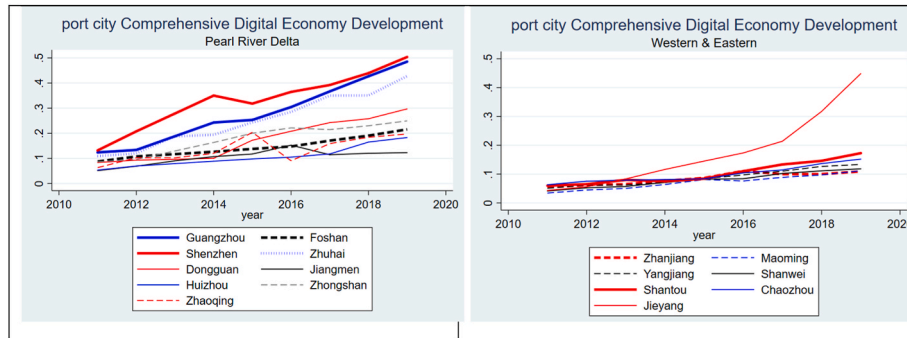
Comprehensive development level of digital economy in Guangdong port cities (Pear River Delta).

Year	Guangzhou	Foshan	Shenzhen	Zhuhai	Dongguan	Jiangmen	Huizhou	Zhongshan	Zhaoqing
2011	0.1241	0.0878	0.1320	0.1093	0.0838	0.0514	0.0537	0.0866	0.0634
2012	0.1338	0.1080	0.2078	0.1205	0.0932	0.0692	0.0699	0.0963	0.1029
2013	0.1884	0.1173	0.2789	0.1879	0.0968	0.0905	0.0795	0.1299	0.1028
2014	0.2429	0.1265	0.3501	0.1941	0.1005	0.1073	0.0891	0.1634	0.1201
2015	0.2533	0.1380	0.3182	0.2435	0.1725	0.1177	0.0980	0.2004	0.2039
2016	0.3045	0.1476	0.3648	0.2854	0.2082	0.1528	0.1052	0.2216	0.0907
2017	0.3668	0.1708	0.3922	0.3502	0.2425	0.1147	0.1180	0.2148	0.1591
2018	0.4270	0.1907	0.4394	0.3507	0.2580	0.1206	0.1650	0.2301	0.1841
2019	0.4853	0.2156	0.5037	0.4276	0.2973	0.1232	0.1832	0.2496	0.1975

Table 6-2

Comprehensive development level of digital economy in Guangdong port cities (western and eastern areas).

Year	Zhanjiang	Maoming	Yangjiang	Shanwei	Shantou	Chaozhou	Jieyang
2011	0.0549	0.0342	0.0509	0.0413	0.0599	0.0630	0.0433
2012	0.0624	0.0449	0.0614	0.0525	0.0650	0.0753	0.0560
2013	0.0650	0.0504	0.0622	0.0569	0.0799	0.0788	0.0832
2014	0.0765	0.0639	0.0707	0.0711	0.0752	0.0811	0.1161
2015	0.0865	0.0812	0.0839	0.0822	0.0841	0.0835	0.1450
2016	0.1107	0.0762	0.0971	0.0839	0.1099	0.1041	0.1734
2017	0.0997	0.0887	0.1102	0.1022	0.1334	0.1145	0.2139
2018	0.1001	0.0970	0.1266	0.1112	0.1460	0.1364	0.3173
2019	0.1091	0.1117	0.1336	0.1183	0.1727	0.1521	0.4486

**Fig. 2.** The Comprehensive Digital Economy development in Guangdong.

Guangzhou, Zhuhai and Jieyang enjoy the highest digital economy development level. And Maoming appears to be the one with the lowest digital economy development level.

4.3. Spatial correlation analysis

The spatial correlation test results are reported in Table 7.

Table 7 reports the spatial autocorrelation coefficients under both the reciprocal distance weight matrix and the economic weight matrix.

For the port city coordination, the spatial autocorrelation coefficients of the port city coupling coordination degrees are basically negative but insignificant under both spatial weight matrices. However,

Table 7

Global Moran's I of port city coordination and digital economy development from 2011 to 2019.

Year	Reciprocal distance weight matrix (Wd)		Economic weight matrix (We)	
	Port city Coupling Coordination degree (port_city_coor)	Digital economy development index (deconomy)	Port city Coupling Coordination degree (port_city_coor)	Digital economy development index (deconomy)
2011	-0.164 (-1.142)	0.149 (2.796)**	-0.138 (-0.344)	0.414 (2.593)***
2012	-0.130 (-0.678)	0.086 (2.383)**	0.046 (0.673)	0.286 (2.297)**
2013	-0.120 (-0.539)	0.017 (1.671)*	-0.223 (-0.786)	0.179 (2.012)**
2014	-0.154 (-0.940)	0.036 (1.538)	-0.324 (-1.278)	0.138 (1.260)
2015	-0.087 (-0.119)	0.093 (2.144)**	-0.263 (-0.954)	0.352 (2.312)**
2016	-0.119 (-0.505)	0.087 (2.064)**	-0.357 (-1.433)	0.320 (2.131)**
2017	-0.148 (-0.868)	0.101 (2.207)**	-0.268 (-0.989)	0.314 (2.063)**
2018	-0.141 (-0.777)	0.049 (1.551)	-0.310 (-1.208)	0.224 (1.586)
2019	-0.153 (-0.950)	0.022 (1.203)	-0.294 (-1.183)	0.144 (1.146)

Note: Z value in brackets, * means significant at 10% level, ** means significant at 5% level, *** means significant at 1% level.

even if the global Moran's I is not significant, the local Moran's I can still be significant, hence further analysis is needed.

For the digital economy development, the global Moran's I of the comprehensive digital economy development levels are positive, and basically significant at 5% level in most of the years from 2011 to 2019, under both spatial weight matrices. It indicates a descent spatial autocorrelation in Guangdong province.

The spatial autocorrelation situation in local areas in more detail can be observed in Figs. 3 and 4 below.

Fig. 3 shows the port city coupling coordination degrees from 2011 to 2019 in Guangdong. The left chart describes the port city coordination in 2011, the middle chart is for 2015 and the right chart is for 2019. The shade of green stands for the degree of port city coordination. Literally, the darker the green, the higher the port city coordination. For instance, in the right chart, as it is in 2019, there are six port cities in the high coupling stage: Guangzhou, Huizhou, Dongguan, Shenzhen, Jiangmen, Zhanjiang, and Shantou.

Overall, the port city coordination in Guangdong represents heterogeneous spatial correlation patterns. In the Pearl River Delta and the western Guangdong area, the coordinated development of port cities show spatial spillover effect, and it seems that a well coordinated port city could drive the coordinated development in the neighborhood. Whereas in the eastern area, the coupling coordination degree shows a spatial agglomeration effect, indicating competition rather than collaboration. These heterogeneous spatial correlation patterns may explain the insignificant global Moran's I for the coupling coordination degrees in Table 7.

Fig. 4 shows the comprehensive digital economy development levels from 2011 to 2019 in Guangdong. The left chart describes the digital economy development in 2011, the middle chart is for 2015 and the right chart is for 2019. The shade of green stands for the levels of digital economy development. Literally, the darker the green, the higher the digital economy development level. For example, in the right chart, as it is in 2019, there are four port cities with the highest level of comprehensive digital economy development: Guangzhou, Shenzhen, Zhuhai, and Jieyang. In general, the comprehensive digital economy development in the Pearl River Delta is higher than those in the eastern and western

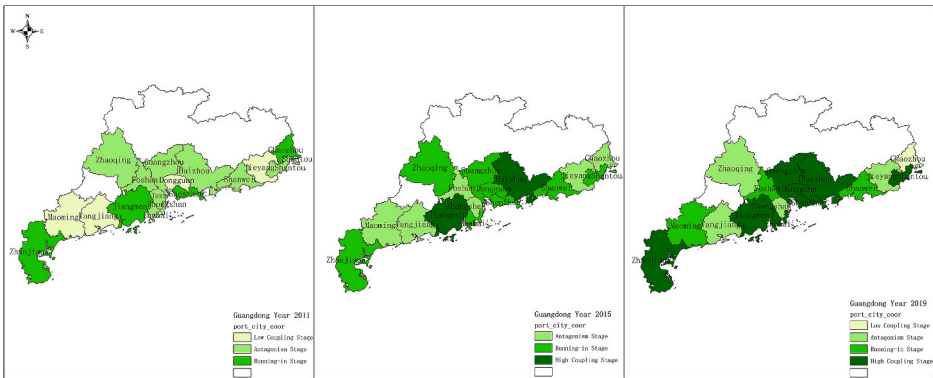


Fig. 3. Port city coupling coordination degree in 2011,2015 and 2019.

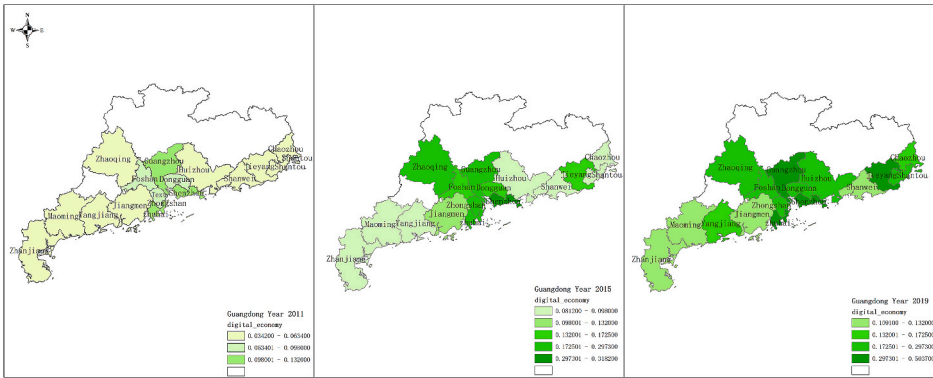


Fig. 4. Comprehensive digital economy development of port cities in 2011,2015 and 2019.

areas. Comparatively, the digital economy development in the eastern area is higher than that in the western area.

Overall, the digital economy development in Guangdong shows positive spatial autocorrelation. In the pearl River Delta, Guangzhou and Shenzhen develop the best comprehensive digital economy. For the cities close to Guangzhou and Shenzhen, the digital development level progress quicker than those cities far away. The same scenario happens in the eastern Guangdong area with Jieyang as the core. As is for the western Guangdong area, the digital economy level develops evenly. The positive spatial autocorrelation is consistent with the results in Table 7.

4.4. Empirical estimation results of spatial model

The estimation results of the dynamic SDM model under two kinds of spatial weight matrices are reported in Table 8.

According to Table 8, under the reciprocal distance spatial weights, the spatial lag term and the temporal lag term of the port city coupling coordination degree play important role in driving the port city coordinated development with their coefficients are both positive and statistically significant at 1% level, indicating that the port city coordination has a positive spatial spill over effect, and that the coupling coordination degrees of port cities have habit persistence. Besides, the coefficient of digital economy development level is significantly positive at the level of 10%, and the coefficient of the spatial lag term of digital economy development is significantly positive at the level of 1%, which suggests that the digital economy development not only improves the local port city coordinated development, but also helps to enhance the port city coordination in the neighborhood. For other explanatory variables, the positive impact of the container throughput and fix asset investment on the development of local port city coordination is significant at the level of 5% and 10% respectively. Meanwhile the

Table 8
Dynamic SDM estimation results.

Explanatory variable	Reciprocal distance weight matrix (Wd)	Economic weight matrix (We)
port_city_coor _{it-1}	0.628 (9.26)***	0.664 (7.76)***
Wport_city_coor _{it-1}	0.493 (1.25)	0.239 (1.37)
Deconomy	0.146 (1.67)*	-0.104 (-1.16)
Lnpdp	0.010 (0.07)	0.027 (0.15)
Lncontainer	0.035 (1.96)**	0.023 (1.47)
Industrystruc	-0.161 (-2.56)***	0.026 (0.37)
Fassetp	0.029 (1.72)*	0.025 (1.29)
Wdeconomy	2.951 (5.19)***	0.547 (2.59)***
Wlnpdp	-0.316 (-0.44)	-0.466 (-2.02)**
Wlncontainer	0.111 (1.25)	0.082 (1.68)*
Windustrystruc	-1.267 (-3.32)***	-0.006 (-0.04)
Wfassetp	0.024 (0.42)	-0.042 (-0.63)
ρ	0.845 (2.83)***	0.138 (0.99)
σ^2	0.002 (7.94)***	0.003 (8.07)***
R ²	0.658	0.632
Log Likelihood	-259.83	140.3329

Note: Z value in brackets, * means significant at 10% level, ** means significant at 5% level, *** means significant at 1% level.

industrial structure has a significant negative impact on both the local and the adjacent port city coordination at the level of 1%.

Under the economic spatial weights in Table 8, the coefficient of the spatial lag term of the port city coupling coordination degree is not significant, but its temporal lag term is significant at the level of 1%, indicating a strong habit persistence and that the local port city coordination can not significantly affect the coordinated development of port cities with similar economic conditions. Besides the coefficient of the spatial lag term of digital economy development is significantly positive at the level of 1%, which shows that the development of digital economy

promotes the port city coordination with similar economic conditions. For other explanatory variables, the coefficient of the spatial lag term of GDP per capita is significantly negative at the level of 5%, suggesting that the economic growth has a negative impact on the coordinated development of port cities with similar economic conditions. Whereas the coefficient of spatial lag term of container throughput is significantly positive at the level of 10%, indicating that the local port operation scale has a positive impact on the coordinated development of port cities with similar economic level.

4.5. Spatial spillover effects of spatial model

Since the coefficients of the spatial lag terms of the independent variables cannot be directly used to explain their marginal impacts, it is necessary to use the partial differential interpretation of the coefficients, so as to decompose the spatial effects of the independent variables on the dependent variable. Specifically for the dynamic SDM model, the spatial effects can be decomposed into short-term direct spatial effects, short-term indirect spatial effects, long-term direct spatial effects and long-term indirect spatial effects. The direct effects are used to explain the impacts of the local independent variables on the local dependent variable, whereas the indirect effects are used to explain the impacts of the local independent variables on the spatially adjacent dependent variable. The decomposition of short-term and long-term spatial effects of the dynamic SDM model are shown in Table 9.

Under the reciprocal distance spatial weights, the direct effects of digital economy development are not significant, but the indirect effects are positive and significant, and the long-term indirect effect is larger than the short-term indirect effect, which shows that the development of the digital economy can promote the coordinated development of the neighboring port cities, and this spatial spillover effect will increase over time. For other variables, the industrial structure have negative and significant indirect spatial effects, and the negative long-term indirect effect is larger than the short-term indirect effect, which suggests that the upgrading of industrial structure hinders the coordinated development of the neighbor's port cities, and that the negative impact will increase over time.

Under the economic spatial weights, the short term indirect spatial effect of the digital economy development level is positive and significant at 1% level, indicating that the development of the digital economy can promote the coordinated development of the port cities with similar economic condition in the short term. For other variables, the short term indirect spatial effect of GDP per capital is negative at the significance level of 10%, which shows that the economic development has negative impact on the coordinated development of port cities with similar economic conditions in the short term. While the short term indirect spatial effect of the container throughput is positive and significant at the level of 10%, suggesting that the improvement of port operation scale can improve the coordinated development of port cities with similar

economic conditions in the short term.

4.6. Spatial spillover effects of sub regions in Guangdong

In order to further clarify the spatial effects of the digital economy development on the coupling coordination degrees of port cities, dynamic SDMs are estimated separately for the Pearl River Delta, and the western and eastern areas according to the geographical division of Guangdong. Since there are relatively few port cities and the economic conditions are similar in the western and eastern Guangdong areas, port cities of the two areas are treated as one group. The estimation results of sub regions are reported in Tables 10-1 and Table 10-2, Tables 10-1 is under the reciprocal distance spatial weights, and Tables 10-2 is under the economic spatial weights.

Under the reciprocal distance spatial weights, both the direct and the indirect spatial effects of the digital economy development in the Pearl River Delta are not significant. In contrast, the indirect spatial effect of the development of digital economy in the western and eastern Guangdong areas is negative and significant, revealing that in the western and eastern areas, the digital economy growth will hinder the coordinated development of neighboring port cities. For other variables,

Table 10-1
Spatial effects of sub regions (Wd).

		Reciprocal distance weight matrix (Wd)			
	Variables	Pearl River Delta		Eastern and Western Guangdong	
		Direct	Indirect	Direct	Indirect
Short Term Spatial Effects	deconomy	-0.187 (-1.48)	0.398 (0.84)	0.006 (0.04)	-1.143*** (-4.15)
	lnpgdp	-0.377* (-1.83)	-0.495 (-0.87)	0.269 (1.46)	0.413 (0.93)
	lncontainer	0.131*** (2.63)	0.163 (0.86)	-0.012 (-1.07)	-0.037 (-1.46)
	Industrystrc	0.429** (2.35)	0.424 (1.53)	0.372 (2.58)	0.811*** (2.82)
	fassetp	0.026 (1.03)	0.006 (0.08)	0.026 (0.29)	-0.371*** (-2.59)
Long Term Spatial Effects	deconomy	0.013 (0.00)	0.257 (0.03)	16.521 (0.06)	-13.074 (-0.04)
	lnpgdp	-0.664 (-0.14)	-0.466 (-0.10)	-6.973 (-0.02)	4.510 (0.01)
	lncontainer	0.435 (0.16)	-0.056 (-0.02)	1.008 (0.05)	-0.844 (-0.04)
	Industrystrc	0.736 (0.21)	0.365 (0.10)	-16.667 (-0.07)	11.240 (0.04)
	fassetp	0.079 (0.04)	-0.039 (-0.02)	6.712 (0.07)	-4.786 (-0.05)

Note: Z value in brackets, * means significant at 10% level, ** means significant at 5% level, *** means significant at 1% level.

Table 9
Analysis of short-term and long-term spatial effects of dynamic SDM.

Independent variables	Short-term spatial effects				Long-term spatial effects			
	Reciprocal distance weight matrix (Wd)		Economic weight matrix (We)		Reciprocal distance weight matrix (Wd)		Economic weight matrix (We)	
	Direct	Indirect	Direct	Indirect	Direct	Indirect	Direct	Indirect
Deconomy	-0.044 (-0.53)	1.623*** (4.16)	-0.124 (-1.37)	0.514*** (2.74)	-0.306 (-0.65)	3.894** (2.32)	0.093 (0.03)	4.541 (0.12)
lnpgdp	0.122 (1.02)	-0.106 (-0.28)	0.042 (0.24)	-0.420 (-1.93)*	0.333 (0.90)	-0.316 (-0.33)	-0.232 (-0.08)	-3.084 (-0.17)
lncontainer	0.027* (1.83)	0.043 (0.89)	0.018 (1.19)	0.074 (1.73)*	0.065 (1.59)	0.093 (0.64)	0.104 (0.22)	0.364 (0.10)
Industrystrc	-0.084 (-1.64)	-0.688*** (-3.24)	0.028 (0.40)	-0.008 (-0.06)	-0.141 (-0.67)	-1.608** (-1.90)	0.062 (0.05)	-0.452 (-0.03)
Fassetp	0.028 (1.55)	0.009 (0.24)	0.027 (1.36)	-0.040 (-0.63)	0.072 (1.42)	0.008 (0.08)	0.088 (0.15)	-0.189 (-0.06)

Note: Z value in brackets, * means significant at 10% level, ** means significant at 5% level, *** means significant at 1% level.

Table 10-2
Spatial effects of sub regions (We).

		Economic weight matrix (We)			
		Pearl River Delta		Eastern and Western Guangdong	
		Direct	Indirect	Direct	Indirect
Short Term Spatial Effects	deconomy	−0.379** (−2.54)	0.904*** (3.17)	−0.479** (−2.28)	−1.003*** (−2.87)
	lnpgdp	−0.448* (−1.94)	−0.522* (−1.72)	−0.604 (−1.54)	1.137** (2.02)
	lncontainer	0.119*** (2.80)	0.096 (1.49)	−0.021 (−1.48)	0.020 (0.60)
	industrystrc	0.096 (0.94)	0.305* (1.87)	0.042 (0.27)	0.972*** (3.2)
	fassetp	0.038* (1.90)	−0.025 (−0.68)	−0.064 (−0.59)	−0.380** (−2.39)
Long Term Spatial Effects	deconomy	−0.761 (−0.28)	2.486 (0.27)	−2.264 (−0.16)	−0.483 (−0.03)
	lnpgdp	−1.348 (−0.51)	−1.741 (−0.09)	1.110 (0.03)	−0.135 (−0.00)
	lncontainer	0.336 (0.82)	0.402 (0.16)	0.067 (0.05)	−0.071 (−0.05)
	industrystrc	0.318 (0.20)	0.778 (0.07)	1.789 (0.07)	0.088 (0.00)
	fassetp	0.091 (1.26)	−0.045 (−0.25)	−0.742 (−0.09)	−0.093 (−0.01)

Note: Z value in brackets, * means significant at 10% level, ** means significant at 5% level, *** means significant at 1% level.

in the Pearl River Delta, the direct effects of the GDP per capita, the container throughput and the industrial structure are positive and significant at 10%, 1% and 5% respectively, which suggests that the economic growth, the port operation scale and the industrial upgrading all contributes positively to the coupling coordination degrees of local port cities. Meanwhile, for the industrial structure, the indirect spatial effect in the western and eastern Guangdong areas is positive and significant at 1% level, showing an active impact of industrial upgrading to the coupling coordination degree of neighbor's port cities. For the fix assets investment, the indirect spatial effect in the western and eastern areas is also significant at 5% level, but negative, indicating that the increased fixed assets investment can suppress the port city coordination in the neighborhood.

Under the economic spatial weights, for the digital economy development, the direct spatial effect is negative and significant at 5% level, and the indirect spatial effect is positive and significant at 1% level in the Pearl River Delta, which indicates that the digital economy growth suppresses the local port city coordination, but prompts the neighboring port city coordination in the Pearl River Delta. In comparison, both the direct and indirect spatial effects in the western and eastern Guangdong areas are negative and significant at 5%, 1% level respectively, which reveals that the digital economy development may hinder the local port city coordination, and the neighboring port city coordination as well in the western and eastern areas.

For other variables, in the Pearl River Delta, the direct spatial effect of the GDP per capita is negative and significant at 10% level, the direct spatial effects of the container throughput and the fixed assets investment are positive, and significant at 1%, 10% level respectively. While the indirect spatial effect of the GDP per capita is negative, significant at 10% level, the indirect spatial effect of the industrial structure is positive, significant at 10% level. Those results suggests that in the Pearl River Delta, the economic growth will affect negatively the coordinated development of both local port cities and port cities with similar economic conditions, and that the port operation scale and the fix assets investment may contribute positively to the local port city coordination, and that the industrial upgrading can promotes the coupling coordination degrees of port cities with similar economic conditions. In contrast, in the western and eastern Guangdong areas, the indirect spatial effects of the GDP per capita and the industrial structure are positive, and significant at 5% and 1% level respectively. For the fix assets investment, the indirect spatial effect is negative and significant at 5% level. It shows that in the western and eastern areas, the GDP growth and the industrial upgrading can improve the coupling coordination degrees of port cities with similar economic conditions, and that the fixed assets investment may influence negatively the coordinated development of port cities in identical economic conditions.

5. Discussion

The port city coupling coordination degrees express the coordinated development of port cities, reflecting sustainable development of port city systems and the regional economy. So far the paper has measured the coupling coordination degrees, discussed the spatial effects of the digital economy development and other factors on the coupling coordination degrees of Guangdong port cities, and found that the port city coupling coordination degrees for different port cities are substantially different.

Generally speaking, the coupling coordination degrees of coastal ports are higher than those of inland ports, and the coupling coordination degrees in the Pearl River Delta are higher than those in the western and eastern Guangdong areas. In the Pearl River Delta, the port city coordination has been growing, and by 2019, all the six coastal ports in the Pearl River Delta have reached the high level of coordination. Whereas in the western and eastern Guangdong areas, the spatial hierarchical structures present a “core-periphery” characteristic, with Zhanjiang and Shantou as the center of western and eastern areas respectively. Zhanjiang is positioned as an important gateway on the west bank of the Pearl River and to the economic circle of the South China Sea. Shantou is positioned as a hub connecting the coastal economic belt and the inland region. Relying on their superior geographical location, dense inland waterway system, dense highway infrastructure and policy support, Zhanjiang and Shantou have gained the highest port city coordination in their respect area.

The digital economy has greatly affected China's social and economic development. In terms of the comprehensive improvement of digital economy, this study has measured the digital economy development level in Guangdong port cities and found that the comprehensive development levels of digital economy in Guangdong cities grow unevenly. Overall, the digital economy develops the best in the Pearl River Delta, less well in the eastern Guangdong, and even weaker in the western Guangdong. Among all the port cities, the digital economy in Jieyang manifested an extraordinary development. Although Jieyang is an inland port in the eastern area, its comprehensive digital economy development level is comparable to the most prosperous cities in the Pearl River Delta. Literally, aside from policies from the national and provincial documents, Jieyang municipal government successively issued the action plans for accelerating the development of the digital economy. Besides, Jieyang has been promoting digital infrastructure, the industrial digital transformation, and its digital economy volume is constantly expanding.

As for the digital economy development pattern, in the Pearl River Delta, the digital economy development in the port cities shows a mode of mutual interactive and drive. Among them, Guangzhou, Shenzhen and Zhuhai play the leading role. In the western Guangdong, the port cities keep a steady and similar growing pace. Whereas In the eastern

area, it shows a spatial agglomeration pattern with Jieyang increasing rapidly and the other port cities remaining stagnant. These structure patterns may result in different policy positioning rather than natural resources or economic conditions.

For the sake of the spatial spillover effects of the digital economy development and other factors on the coordinated development of port cities, the study has found that in the estimation of spatial model, the coefficients of digital economy turns out to be significant, and the indirect spatial effects are greater than the direct spatial effects. It indicates that the digital economy can promote the port city coordination where spatially adjacent, greater than the digital economy can improve the coordinated development in the local area. Furthermore, the spatial spillover effects of digital economy are substantially different for different regions in Guangdong.

In the Pearl River Delta, when is calculated under the economic spatial weights, the direct effect of the digital economy is negative and significant, the indirect effect is positive and significant, suggesting that the development of digital economy may hinder the coordinated development of local port cities, but improve the coordinated development of economically adjacent port cities. Meanwhile, In the western and eastern Guangdong areas, when is calculated under the reciprocal distance spatial weights, the direct effect of the digital economy is insignificant, but the indirect effect is negative and significant, showing that a port city which enjoys high digital economy development does not necessarily gain a high coordinated development, and that a port city with high comprehensive digital economy development may negatively influence the coordinated development of neighboring port cities.

According to the spatial empirical analysis, there are other elements which also impact the coupling coordination degrees of port cities. Firstly, the temporal lag term of the coupling coordination degree is considered. The research found that no matter under the reciprocal distance spatial weights or under the economic spatial weights, no matter is for the Guangdong province or is for the sub regions, the coefficients of the temporal lag term of the coupling coordination degree are positive and significant, representing a strong habitat persistence, which means that the past coupling coordination degree will have a great positive impact on the current coupling coordination degree.

Secondly, for the GDP per capita, when is estimated for the whole province, a negative indirect effect is observed under the economic spatial weights, which suggests a negative impact of economic growth on the coordinated development of port cities with similar economic conditions. It shows possible competition among ports with similar economic conditions, and the homogeneous competition can influence the port city coordination negatively. When the GDP per capita is estimated for the sub regions, in the Pearl River Delta, both the direct and the indirect spatial effects are negative and significant under the economic spatial weights, indicating that the city economic scale contribute negatively both to its local port city coordination and to its economically adjacent port cities coordination. For the western and eastern areas, the indirect spatial effect is positive and significant under the economic spatial weights, showing that the city economic scale contributes positively to the coordination development of its economically adjacent port cities. The entirely different estimation results in the Pearl River Delta and the western and eastern areas suggest a more severe competition among ports with similar economic conditions in the Pearl River Delta, and a possible cooperation among ports with similar economic conditions in the western and eastern areas.

Then for the container throughput, a positive and significant direct spatial effect is observed in the estimation of the whole province under the reciprocal distance spatial weights, suggesting a positive impact of port operation scale on the coordinated development of the local port city. Referring to the sub regions regression, in the Pearl River Delta, under both the reciprocal distance spatial weights and the economic spatial weights, the container throughput has a positive and significant direct spatial effect and insignificant indirect spatial effect. In the western and eastern areas, both the direct and indirect spatial effects are

insignificant. The results show that the impact of the port operation scale is positive on the local port city coordination in the Pearl River Delta, whereas although the overall impact is positive, the impact of the container throughput on the coordinated development in the western or eastern area is not so obvious.

For the industrial structure, a negative short term indirect effect and a negative long term indirect effect on the port city coordination are detected under the reciprocal distance spatial weights, significant at 1% and 5% level respectively. The results shows that the industrial upgrading impact negatively the neighboring port city coordination. Referring to the sub regions regression results, In the Pearl River Delta, under the reciprocal distance spatial weights the industrial structure has a positive and significant direct spatial effect, and under the economic spatial weights, the industrial structure has a positive and significant indirect spatial effect. In the western and eastern areas, under both the reciprocal distance spatial weights and the economic spatial weights, the industrial structure has a positive and significant indirect effect. The results show that the industrial upgrading may improve the local port city coordination, and can prompt the coordinated development of economically adjacent port cities as well in the Pearl River Delta. In contrast, in the western and eastern areas, the industrial upgrading can exert positive impact on the coordinated development of both geographically and economically adjacent port cities.

For the fixed asset investment, referring to the sub regions regression results, in the Pearl River Delta, the fixed asset investment has a positive and significant direct spatial effect under the economic spatial weights. In the western and eastern areas, under both the reciprocal distance spatial weights and the economic spatial weights, the fixed asset investment has a negative and significant indirect effects. The results reveal that the fixed asset investment increase may prompt the local port city coordination in the Pearl River Delta, and can suppress the coordinated development of both geographically and economically adjacent port cities.

In sum, all the empirical evidences reveal that the digital economy development is helpful to form an economic pattern of coordinated development of port clusters. It also brings the policy implications as follows:

First of all, the development of digital economy can promote the port city coordination, therefore it is suggested to increase the investment in the Internet and to promote digital construction, especially by accelerating 5G commercial use, big data model construction and artificial intelligence application, to further consolidate the advantage of information technology for the regional economic integration of the port and city. Secondly, the spatial spillover effect of the digital economy shows that the coordination of new digital infrastructure construction should be strengthened to fully release its spatial contribution to port city coordination. In addition, the regional individual fixed effect indicates that the development of port city coupling coordination degree in Guangdong is not balanced, which indicates that a dynamic and differentiated digital economy strategy can be implemented, making the digital economy a “hardware” technical support to effectively reduce the imbalance of regional development.

6. Conclusions

Ports are regarded as an important transportation infrastructure and play a key role in the hinterland economic development. The digital economy has greatly affected China's social and economic development, intelligent logistics, and that cross-border e-commerce has promoted the development of the fifth generation of smart ports. However, the evaluation of the impact of the digital economy development on the port city coordination, and its spatial effects are seldom quantitatively investigated, especially in the context of Guangdong being a densely allocated coastal and inland port group.

Based on the data from Guangdong where coastal and inland port clusters are densely allocated and possessed high volume of digital

economy, in this paper, we measures the coupling coordination degrees of port cities by the coupling coordination model, calculates the comprehensive digital economy development levels by the TOPSIS entropy weight method, selects panel data of port cities in Guangdong from 2011 to 2019, and constructs the dynamic SDM, respectively uses the reciprocal distance weight and economic weight matrices, empirically tests the spatial effect of digital economy on the port city coordination. Finally, applies the dynamic SDM on the sub regions of Guangdong to further analyze the spatial effects of the digital economy on the port city coordination within and between sub regions. The main findings are as follows:

First, in terms of the port city coordinated development, port cities in Guangdong varies with distinct individual characteristics. In general, the coupling coordination degrees in the Pearl River Delta are higher than those in eastern and western areas, and the coupling coordination degrees of sea ports are better than those of inland ports. Also, it is observed that the port city coordination has a habit persistence, that is, the past port city coordination will contribute positively to the future port city coordination.

Second, with regard to digital economy, the comprehensive development has an increasing tendency in all Guangdong cities. And it is concluded that the development of digital economy may be more related to the local policy than to the regional economy advantages.

Third, in light of the spatial spill over effects, the development of the digital economy can promote the coordinated development of the economically adjacent port cities more than it does to the local port cities, and this spatial spillover effect will increase over time. Moreover, different patterns of the spatial spill over effects of digital economy development on the coordinated development of port cities exist among the sub regions of Guangdong.

Fourth, in terms of others factors on the port city coordinated development, the industrial structure upgrading and economy development results a negative impact on the neighboring port city coordination, which indicates competition among port cities. Whereas the container throughput exerts a positive impact on the port city coordination in the neighborhood, which indicates possible logistics cooperation among port cities.

The research of this paper has the following limitations: the investigation of the spatial effects of the digital economy development does not control the impact of the temporal lag term of the digital economy development and the freight turnover factor. The freight turnover factor and the temporal lag term of the digital economy development may prove fruitful.

CRediT authorship contribution statement

Bo Lin: Conceptualization, Supervision, and, Writing – review & editing. **Jie Tang:** Conceptualization, Methodology, and, Writing – original draft. **Chang Dai:** Methodology, Validation, and, Writing – original draft. **Bo Lu:** Methodology, Supervision, and, Writing – original draft.

Declaration of competing interest

On behalf of all authors, no potential conflict of interest was reported by the authors.

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References

- A, G., S, K., S, M., & R, M. (2021). Port and city integration: Transportation aspect. *Transportation Research Procedia*, 54, 890–899. <https://doi.org/10.1016/j.trpro.2021.02.144>
- Chen, C., & Song, F. F. (2019). Evaluation of port logistics competitiveness in China along the belt and road. *Journal of Coastal Research*, 93, 1117–1124.
- Cheng, Y., Zhang, Y., Wang, J., & Jiang, J. (2023). The impact of the urban digital economy on China's carbon intensity: Spatial spillover and mediating effect. *Resources, Conservation and Recycling*, 189, 106762.
- Cong, L. Z., Zhang, D., Wang, M. L., Xu, H. F., & Li, L. (2020). The role of ports in the economic development of port cities: Panel evidence from China. *Transport Policy*, 90, 13–21.
- Dadashpoor, H., & Arasteh, M. (2020). Core-port connectivity: Towards shaping a national hinterland in a west Asia country. *Transport Policy*, 88(1), 57–68.
- Deng, Z., L, Z.F., Zhou, Y.T., C, Xiao., & L, S. S. (2020). Measurement and spatial spillover effects of port comprehensive strength: Empirical evidence from China. *Transport Policy*, 99, 288–298. doi:10.1016/j.tranpol.2020.09.006.
- Deng, H., Bai, G., Shen, Z., & Xia, L. (2022). Digital economy and its spatial effect on green productivity gains in manufacturing: Evidence from China. *Journal of Cleaner Production*, 378(2022), 134539.
- Ducruet, C., Cuyala, S., & El Hosni, A. (2018). Maritime networks as systems of cities: The long-term interdependencies between global shipping flows and urban development (1890–2010). *Journal of Transport Geography*, 66(1), 340–355.
- Elhorst, J. P. (2014). Matlab software for spatial panels. *International Regional Science Review*, 37(3), 389–405.
- Feng, H., Grifoll, M., & Zheng, P. (2019). From a feeder port to a hub port: The evolution pathways, dynamics and perspectives of Ningbo-Zhoushan port (China). *Transport Policy*, 76(2), 21–35.
- Feng, G. F., Wang, Q. J., Wen, J., Dong, M., & Chang, C. P. (2018). Do economic growth and seaport throughput move together cities. *International Journal of Transport Economics*, 45(2), 211–239.
- G, D. A., K, S. D., I, D., & G, L. (2021). Smart and sustainable logistics of port cities: A framework for comprehending enabling factors, domains and goals. *Sustainable Cities and Society*, 69, Article 102801. <https://doi.org/10.1016/j.scs.2021.102801>
- Guo, J., Qin, Y., Du, X., & Han, Z. (2020). Dynamic measurements and mechanisms of coastal port-city relationships based on the DCI model: Empirical evidence from China. *Cities*, 96, 102440.
- Guo, J. K., & Qin, Y. F. (2022). Coupling characteristics of coastal ports and urban network systems based on flow space theory: Empirical evidence from China. *Habitat International*, 126, Article 102624. <https://doi.org/10.1016/j.habitatint.2022.102624>
- Han, E. Y., Wang, D. R. N., & Li, B. (2019). Spillover effects of ports and logistics development on economic power: Evidence from the Chinese BTH regions. *Sustainability*, 11(8), 1–17.
- He, D., Sun, Z., Gao, P., & Lau, Y. Y. (2019). Spatial-temporal evolution of the port-hinterland relationship: A case study of the midstream yangtze river, China. *Growth and Change*, 50(3), 1043–1061. <https://doi.org/10.1111/grow.12320>
- Hou, Y., Long, R., Zhang, L., & Wu, M. (2020). Dynamic analysis of the sustainable development capability of coal cities. *Resources Policy*, 66, Article 101607. <https://doi.org/10.1016/j.resourpol.2020.101607>
- Kong, Y. D., & Liu, J. G. (2021). Sustainable port cities with coupling coordination and environment efficiency. *Ocean & Coastal Management*, 205, Article 105534. <https://doi.org/10.1016/j.ocecoaman.2021.105534>
- Lam, J. S. L., & Li, K. X. (2019). Green port marketing for sustainable growth and development. *Transport Policy*, 84, 73–81. <https://doi.org/10.1016/j.tranpol.2019.04.011>
- Li, X. K., Li, M. C., Zhu, Y. H., et al. (2023). Smart port: A bibliometric review and future research directions. *Transportation Research Part E: Logistics and Transportation Review*, 174, Article 103098. <https://doi.org/10.1016/j.tre.2023.103098>
- Lim, S., Pettit, S., Abouarghoub, W., & Beresford, A. (2019). Port sustainability and performance: A systematic literature review. *Transportation Research Part D: Transport and Environment*, 72, 47–64. <https://doi.org/10.1016/j.trd.2019.04.009>
- Lin, B., & Tang, J. (2023). Impact of port-city coordination upon hinterland economy in Pearl River Delta. *International Journal of Innovative Computing, Information and Control*, 19(1), 289–297.
- Lee, T. W., Lam, J., Lin, C. W., Hu, K. C., & Cheong, I. (2018). Developing the fifth generation port concept model: An empirical test. *International Journal of Logistics Management*, 29(3), 1098–1120.

- Li, B., Shi, Z. Y., & Tian, C. (2018). Spatial-temporal difference and influencing factors of environmental adaptability measurement of human-sea economic system in Liaoning coastal area. *Chinese Geographical Science*, 28(2), 313–324.
- Liu, Z. G., Schindler, S., & Liu, W. D. (2020). Demystifying Chinese overseas investment in infrastructure: Port development, the Belt and Road Initiative and regional development. *Journal of Transport Geography*, 87, Article 102812. <https://doi.org/10.1016/j.jtrangeo.2020.102812>
- Lu, B., Qiu, W. Q., Xing, J., & Wen, Y. J. (2020). Research on the development strategy of ports and port cities in China's coastal nodes based on the evaluation of the "Belt and Road" initiative. *Systems Engineering-Theory & Practice*, 40(6), 1627–1639.
- Lu, B., Xing, J., Wang, Q., & Wang, S. Y. (2019). Panel data analysis of port competitiveness and hinterland economic synergy mechanism. *Systems Engineering-Theory & Practice*, 39(4), 1079–1090.
- Mudronja, G., Jugovi, A., & Skalamera-Alilovi, D. (2020). Seaports and economic growth: Panel data analysis of EU port regions. *Journal of Marine Science and Engineering*, 8(12), 1017.
- Pan, W., Xie, T., W, Z. W., & Ma, L. (2022). Digital economy: An innovation driver for total factor productivity. *Journal of Business Research*, 139, 303–311. <https://doi.org/10.1016/j.jbusres.2021.09.061>
- Sdoukopoulos, E., & Boile, M. (2020). Port-hinterland concept evolution: A critical review. *Journal of Transport Geography*, 86, Article 102775. <https://doi.org/10.1016/j.jtrangeo.2020.102775>
- Shi, D. (2022). The evolution of industrial development trend under the condition of digital economy. *China Industrial Economics*, 11, 2642. <https://doi.org/10.19581/j.cnki.ciejournal.2022.11.001>
- Tang, J., & Lin, B. (2023). Ports and economic development of port cities in Pearl River Delta based on Panel Vector Autoregressive Model. *International Journal of Innovative Computing, Information and Control*, 19(3), 835–843.
- Wang, J., & Deng, Y. (2020). Research on the coupling and coordination of port logistics and direct hinterland economy: A case study of Tianjin, yingkou and other nine-port national logistics hubs. *Industrial Technological Economics*, 39(11), 62–68.
- Xu, Z., Gao, Y., & Huo, Z. F. (2022). The impact of digital economy development on urban economic agglomeration. *Statistics & Decisions*, 38(14), 80–84. <https://doi.org/10.13546/j.cnki.tjjyc.2022.14.016>
- Zhao, T., Zhang, Z., & Liang, S. K. (2020). Digital economy, entrepreneurial activity and high-quality development: Empirical evidence from Chinese cities. *Management World*, 36(10), 65–76. <https://doi.org/10.19744/j.cnki.11-1235/f.2020.0154>